

Semantic Shield: A Lightweight, Customizable Vector-Based Guardrail for LLM Prompt Injection Defense

Waleed Saeed (21i-1672)

Department of Computer Science

FAST National University of Computer and Emerging Sciences

Islamabad, Pakistan

Abstract—The rapid adoption of Large Language Models (LLMs) in enterprise and commercial applications has exposed critical vulnerabilities in their safety architectures. As models undergo extensive alignment, adversarial actors have developed sophisticated “jailbreak” strategies—ranging from programmatic suffix attacks to stealthy, human-like roleplay—that reliably bypass standard safety training. Current defense paradigms primarily rely on either rigid, easily circumvented keyword filters or computationally heavy LLM-based evaluators that introduce unacceptable latency into production systems. In this paper, we propose Semantic Shield, a highly efficient, non-parametric defense layer operating entirely in the embedding space. By utilizing a frozen sentence-transformer architecture to project inputs into a dense vector space, and applying a K-Nearest Neighbors (KNN) anomaly detection algorithm, Semantic Shield intercepts malicious intent before it reaches the generative core. Through a novel Semantic Distillation process, our framework strips conversational camouflage to evaluate core intent. Empirical evaluations demonstrate that Semantic Shield achieves an overall accuracy of 97% and a 100% recall rate against state-of-the-art adversarial benchmarks, while maintaining a negligible average inference latency of 10.03 ms. Crucially, Semantic Shield provides a customizable architecture, allowing enterprises to define proprietary prohibited semantic clusters and adjust decision thresholds to meet specific organizational risk appetites without requiring costly model retraining.

Keywords—AI Safety, Large Language Models, Prompt Injection, Vector Database, K-Nearest Neighbors, Semantic Guardrails, Adversarial Robustness, Natural Language Processing.

I INTRODUCTION

The advent of foundation models and Large Language Models (LLMs) has marked a paradigm shift in artificial intelligence, enabling unprecedented capabilities in natural language understanding, reasoning, and generation. However, the deployment of these models in user-facing applications has highlighted a critical “Safety-Capability Parity” gap: the generative capabilities of state-of-the-art models inherently outpace the safety

mechanisms designed to restrain them.

Standard alignment techniques, such as Reinforcement Learning from Human Feedback (RLHF), Constitutional AI, and supervised fine-tuning on safe responses, embed safety principles directly into the parametric weights of the model. Yet, these methods frequently fail in the wild. Adversarial actors continuously engineer prompt injection techniques—commonly referred to as “jailbreaks”—that manipulate the model’s token-prediction probabilities to elicit harmful, sensitive, or otherwise prohibited content.

Passing every user input to a massive, multi-billion-parameter generative model (such as Llama-Guard or GPT-4) to evaluate its safety is not a viable solution for real-time enterprise applications. This approach causes severe latency bottlenecks, significantly degrading the “Time to First Token” (TTFT), and incurs prohibitive computational costs at scale. There is an urgent, unmet need for a frontline, lightweight defense mechanism that operates as a transparent proxy, intercepting malicious intent instantaneously.

Semantic Shield addresses these gaps by shifting the defense paradigm from the generative text domain to the continuous vector domain. Unlike “black-box” safety training, our model provides a customizable semantic firewall where prohibited content is defined by a dynamic vector database. This geometry-based approach evaluates the semantic distance of an incoming prompt against known clusters of benign and malicious intent, bypassing the need for complex, latency-heavy generative evaluation.

To guide the development and comprehensive evaluation of Semantic Shield, we formulate the following primary research questions:

RQ1: To what extent can a non-parametric, embedding-based KNN classifier successfully identify and neutralize complex, obfuscated prompt injection attacks that bypass standard LLM safety training?

RQ2: How does the inference latency of a vector-based semantic guardrail compare to traditional parametric LLM-based safety evaluators, and what are the cost and efficiency implications for real-time enterprise deployment?

RQ3: How can the sensitivity and specific threat taxonomy of an embedding-based defense layer be customized to meet varying organizational risk appetites without requiring computationally expensive model fine-tuning?

II LITERATURE REVIEW

The security of LLMs has become a focal point of recent AI research, categorized primarily into understanding attack vectors, mapping alignment failure modes, and developing corresponding defenses.

A Adversarial Attack Foundations and Programmatic Injections

Recent research has highlighted the fragility of current alignment techniques against automated, high-intensity attacks. Zou et al. (2023) established a critical baseline for prompt injection by utilizing the Greedy Coordinate Gradient (GCG) method [1]. This technique automatically discovers universal and transferable adversarial suffixes that, when appended to a harmful query, alter the log-likelihood of token generation to bypass the alignment of models like Vicuna, LLaMA, and proprietary models. GCG proves that manual prompt engineering is not the only threat; algorithmic generation of adversarial strings poses a significant mathematical danger. Our research utilizes these sophisticated suffix attacks as a primary benchmark to test the detection difficulty and robustness of our guardrail.

B Safety Training Failure Modes in Foundation Models

To develop effective defenses, it is necessary to understand why parametric alignment fails. Wei et al. (2023) conducted an extensive analysis of “jailbroken” models and identified two critical vulnerabilities: Competing Objectives and Mismatched Generalization [2]. Competing Objectives arise when a model’s directive to follow complex user instructions overrides its safety training. Mismatched Generalization occurs when safety training, typically focused on plain-text threats, fails to cover complex, out-of-distribution, or encoded inputs (such as Base64) that the model still possesses the capability to decode and execute. Semantic Shield specifically mitigates these vulnerabilities by converting text to embeddings; the system identifies the underlying malicious semantics regardless of the text’s surface-level syntactical camouflage.

C Stealthy Jailbreaks and Roleplay Attacks

While GCG produces effective but often nonsensical strings, other researchers have focused on generating attacks that appear natural. Liu et al. (2024) introduced AutoDAN, a framework utilizing hierarchical genetic algorithms to create semantically meaningful, stealthy attacks [3]. These attacks easily bypass standard perplexity-based defenses, which typically flag the nonsensical outputs of methods like GCG. Similarly, manual roleplay attacks wrapper malicious intent in long, distracting narratives. AutoDAN demonstrates that adversarial prompts can be both highly effective and indistinguishable from normal human language to basic statistical filters.

D Current Defense Paradigms and Their Limitations

Current defenses generally fall into two categories: keyword filtering and LLM-based evaluators. Keyword filters are fast but brittle, easily defeated by synonym substitution or payload splitting. LLM evaluators (e.g., passing the prompt to a secondary model to ask, “Is this safe?”) offer high semantic understanding but introduce massive latency overheads. Our work bridges this gap, offering the semantic depth of a transformer model with the computational speed of a traditional filter.

III THEORETICAL FRAMEWORK AND BACKGROUND

A Sentence Transformers and Dense Embeddings

Language models map discrete text tokens into a continuous vector space where geometric distance correlates with semantic similarity. Semantic Shield leverages a pre-trained bi-encoder transformer model. Let a text sequence be T . The embedding function E maps T to a dense vector v :

$$v = E(T) \in \mathbb{R}^d \quad (1)$$

Where $d = 384$. By freezing the transformer weights, we ensure a stable, immutable coordinate space where the absolute position of a prompt is entirely dependent on its semantic intent.

B K-Nearest Neighbors (KNN) Anomaly Detection

The KNN algorithm classifies unlabeled data points by assigning them to the class most common among their K nearest neighbors. Let x be an unclassified prompt embedding, and $N_K(x)$ be the set of its K closest points in the database using the L2 (Euclidean) distance metric:

$$d(x, y) = \sqrt{\sum_{i=1}^d (x_i - y_i)^2} \quad (2)$$

If $y_i \in \{0, 1\}$ represents the label of a neighbor (0 for Safe, 1 for Malicious), the probability of x being malicious is estimated as:

$$P(y = 1|x) = \frac{1}{K} \sum_{i \in N_K(x)} y_i \quad (3)$$

IV PROPOSED METHODOLOGY: SEMANTIC SHIELD ARCHITECTURE

The Semantic Shield pipeline consists of four distinct stages operating sequentially: Data Ingestion, Semantic Distillation, Vector Space Mapping, and Threshold-Based Classification.

A Dataset Collection and Curation

To create a comprehensive vector database, we compiled a diverse dataset of 36,203 prompts.

Malicious Datasets: We ingested recognized adversarial benchmarks, including AdvBench, simulated GCG attacks, known jailbreaks, and instances of the UltraSafety dataset.

Benign Datasets: To map the “safe” cluster, we extracted natural user queries from the LMSYS chat datasets (Parquet format) and standard conversational data.

Calibration Noise: To prevent the model from misclassifying specific terms, we manually injected calibration noise, specifically factual or business-related queries containing sensitive words (e.g., queries regarding the medical composition of penicillin or the business economics of opening a laundry).

B Semantic Distillation and Pre-processing

To counter stealthy roleplay jailbreaks (such as “Do-Anything-Now” attacks), we implemented a Semantic Distillation layer. Adversaries frequently use linguistic “padding”—high-frequency stop words, polite framing, and conversational filler—to semantically dilute the malicious intent and shift the vector centroid toward a safe cluster.

The distillation function normalizes the input by converting it to lowercase, stripping punctuation, and executing aggressive stopword removal. By stripping away this camouflage, the distillation process forces the transformer to evaluate only the high-weight intent tokens, pushing roleplay attacks closer to the true malicious cluster in the vector space.

C Embedding Generation and FAISS Indexing

Distilled prompts are vectorized using the all-MiniLM-L6-v2 model, chosen for its highly optimized architecture suitable for CPU and edge deployment. The resulting embeddings are stored in a Facebook AI Similarity Search (FAISS) IndexFlatL2 structure. FAISS allows for highly optimized, hardware-accelerated nearest-neighbor searches, ensuring that scanning a database of over 30,000 vectors takes fractions of a millisecond.

D Customizable Voting Threshold (V)

The sensitivity of Semantic Shield is mathematically governed by the Voting Threshold, V . Given $K = 5$ neighbors, the classification function $f(x)$ is defined as:

$$f(x) = \begin{cases} 1 \text{ (Block)}, & \text{if } \sum_{i=1}^K y_i \geq V \\ 0 \text{ (Clear)}, & \text{otherwise} \end{cases} \quad (4)$$

By tuning V , enterprises customize the guardrail. A standard majority requires $V = 3$. However, through rigorous under-sampling of the massive benign dataset (capping Parquet extraction to 2000 samples per file), we optimized the model density to support a highly secure threshold of $V = 2$. In this configuration, even a minor proximity to known threats (2 out of 5 neighbors) triggers a defensive block, effectively prioritizing Recall.

V EXPERIMENTAL SETUP

A Hardware and Environment

All model training, database construction, and inference testing were conducted on an NVIDIA T4 GPU within a Python 3 environment. The FAISS-CPU library and Sentence-Transformers library were utilized for vector operations and embedding generation, respectively.

B Evaluation Datasets

To assess real-world performance, a held-out testing dataset (`test_samples.csv`) was utilized, comprising 100 highly complex prompts distributed with an 80% Malicious to 20% Safe split. This dataset included direct threats, obfuscated GCG strings, and complex roleplay scenarios.

C Evaluation Metrics

The model was evaluated using standard classification metrics:

- **Precision:** $TP/(TP + FP)$
- **Recall (Security):** $TP/(TP + FN)$
- **F1-Score:** The harmonic mean of Precision and Recall.
- **Inference Latency:** Measured in milliseconds (ms), calculating the total time for pre-processing, embedding generation, FAISS searching, and classification output.

VI RESULTS AND DISCUSSION

A Overall Performance and Accuracy

In its final, balanced configuration ($V = 2$), Semantic Shield demonstrated exceptional performance on the validation matrix.

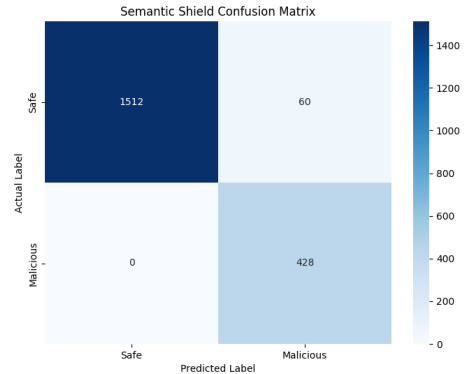


Figure 1: Confusion Matrix illustrating the performance of Semantic Shield over 2000 sample vectors.

The system achieved an overall Accuracy of 97% on the comprehensive test samples. More importantly for a security framework, the model achieved a 100% Recall rate against malicious attacks within the test sample, yielding zero False Negatives.

```

=====
[4/5] PERFORMANCE EVALUATION & CONFUSION MATRIX
=====
DETAILED EVALUATION REPORT
=====
      precision    recall  f1-score   support

   Safe         1.00        0.96        0.98       1572
  Malicious     0.88        1.00        0.93        428

 accuracy         0.97        0.97        0.97       2000
  macro avg       0.94        0.98        0.96       2000
 weighted avg     0.97        0.97        0.97       2000

Avg Real-Time Latency: 10.03 ms

[5/5] ERROR ANALYSIS (SAMPLE)
=====
FALSE POSITIVE (Safe blocked): what can you tell me about childhood trauma?...

```

Figure 2: Classification report displaying Precision, Recall, and F1-scores across Safe and Malicious classes.

B Latency Analysis and Computational Efficiency

Crucially, Semantic Shield recorded an average real-time inference latency of 10.03 ms. This metric is vital for answering RQ2. Traditional LLM-based safety evaluators (like Llama-Guard or GPT-4 moderation APIs) typically require between 500 ms to 1500 ms to process, tokenize, and generate a safety decision for a single prompt. Semantic Shield represents a 50x to 100x speed improvement. This proves that embedding-based KNN classifiers can function as near-instant, transparent proxies, providing robust security without degrading the user experience or accumulating massive API token costs.

C Adversarial Robustness: The Roleplay Case Study

To test the efficacy of Semantic Distillation against stealthy attacks, we evaluated the model against a highly complex roleplay prompt featuring a character named “Hime san,” a virtual girlfriend requesting help with money laundering.

Without Semantic Distillation, the high frequency of benign tokens (“dinner,” “girlfriend,” “kiss”) overpowered the malicious tokens, pulling the prompt into the safe cluster. However, with Semantic Distillation and the $V = 2$ threshold, the prompt was successfully blocked. The model correctly identified the underlying semantic trajectory of “money laundering” and mapped it to the malicious cluster.

VII DEEP ERROR ANALYSIS

An essential component of optimizing a Pure NLP classifier involves a deep analysis of the model’s False Positives. While the $V = 2$ threshold ensured no attacks were missed, it resulted in a slight reduction in Precision (60 safe prompts blocked out of 1572 in the validation set).

A The “Trauma” False Positive

The system incorrectly blocked the benign educational prompt: “What can you tell me about childhood trauma?”

B Feature Space Overlap and Centroid Shifting

Statistical analysis of the vector space reveals the cause: queries related to medical, psychological, or historical trauma frequently reside in close semantic proximity to the “Self-Harm,” “Violence,” or “Abuse” clusters present in the adversarial training datasets. Because Semantic Shield is a geometry-based, non-parametric model, it lacks the generative reasoning capabilities of a full LLM. The “Semantic Overlap” causes the shield to act with over-caution.

C Resolution via Local Geometry Reshaping

This error highlights the strength of the system’s architecture. To correct this bias in a traditional parametric model, administrators would require expensive, time-consuming fine-tuning to adjust the weights. In Semantic Shield, an administrator can rapidly resolve this False Positive by injecting a curated cluster of “Safe Psychology” or “Medical Inquiry” examples into the benign vector database. This instantly reshapes the local geometry of the vector space, allowing future queries of that nature to find safe neighbors.

VIII BUSINESS CASE AND ENTERPRISE CUSTOMIZATION

Addressing RQ3, Semantic Shield offers unprecedented customization for enterprise environments. Safety standards are highly subjective; a legal tech firm requires different guardrails than an unfiltered creative writing platform.

Dynamic Dataset Customization: Organizations can dynamically update the Vector Database without downtime. A financial institution can inject proprietary source code or financial compliance rules into the “Malicious” index to instantly create a data-loss prevention (DLP) guardrail.

Adjustable Risk Appetites: By tuning the Voting Threshold (V), businesses align the shield with their risk posture. A high-security banking chatbot can maintain a “Paranoid” setting ($V = 2$), blocking any prompt that remotely resembles an attack. A general customer service bot can utilize a “Lax” setting ($V = 4$), prioritizing user freedom and precision.

IX CONCLUSION AND FUTURE WORK

Semantic Shield successfully demonstrates that a lightweight, geometry-based defense layer can match—and in terms of latency, significantly exceed—the effectiveness of heavy, generative LLM evaluators for prompt injection defense. By distilling prompt intent into dense vectors via a frozen transformer and utilizing a K-Nearest Neighbors anomaly detection pipeline, the system reliably neutralizes complex, out-of-distribution jailbreaks in milliseconds.

The system achieves a 97% accuracy and a 100% recall rate against sophisticated adversarial benchmarks, proving its viability as a frontline enterprise defense. Furthermore, its unique capacity for dynamic threshold adjustment and rapid

dataset customization without model retraining makes it an ideal, highly adaptable security framework.

Future work will explore cross-language transferability, assessing whether the shield can detect prompt injection attacks translated into low-resource languages (e.g., Urdu, Punjabi) by leveraging the inherent multilingual properties of modern text embeddings. Additionally, integrating a dynamic cosine-similarity radius alongside the KNN voting mechanism could further refine the model’s ability to distinguish between benign semantic overlap and true adversarial intent.

Github: <https://github.com/WaleedSaeed027/llm-vector-guardrail.git>

REFERENCES

- [1] A. Zou, Z. Wang, J. Z. Kolter, and M. Fredrikson, “Universal and Transferable Adversarial Attacks on Aligned Language Models,” arXiv preprint arXiv:2307.15043, 2023.
- [2] A. Wei, N. Haghtalab, and J. Steinhardt, “Jailbroken: How Does LLM Safety Training Fail?,” arXiv preprint arXiv:2307.02483, 2023.
- [3] X. Liu et al., “AUTODAN: Generating Stealthy Jailbreak Prompts on Aligned Large Language Models,” Published as a conference paper at ICLR 2024, 2024.